

Element C Report

Presentation and Justification of Solution Requirements

Problem Statement

Reports indicate that 42,000 American homeowners annually sustain recurring neck, shoulder, and back injuries while picking up and disposing of leaves in their yards. Disregarding this task causes leaves to accumulate in yards, smothering grass and blocking the sunlight needed for proper growth. Overall, leaf removal is a task that can cause physical strain when done and, if neglected, can affect the yard's health, leading to grass death in the fall.

Introduction

Design requirements are created when decisions are being made towards a solution to a problem. These requirements define how effective and unique the solution shall be and ensure it matches the customer's needs. Below are the requirements for the problem of physical strain while picking up leaves from the lawn.

Primary Stakeholders

- Homeowners who maintain their lawns are most directly impacted by the physical demands and time required for leaf removal.
- Older individuals with physical limitations Face difficulties in performing physically demanding tasks like leaf removal and put themselves at risk of serious injury.
- Landscaping service providers Require efficient tools to increase productivity and time spent on the task
- Homeowners Associations (HOA)

Design Requirements

1. Autonomous Operation
 - a. Specification:
 - i. The machine shall operate fully autonomously once it is set up and trained
 - b. Rationale:
 - i. Automation is critical to minimizing physical intervention, reducing the risk of injury, and strain.

2. Efficiency

a. Specification:

- i. The machine shall pick up leaves from a 2,500 sq. ft. lawn within 60 minutes under weather conditions such as clear sky or some wind.

b. Rationale:

- i. This ensures that the machine performs at a speed comparable to manual raking or blowing, making the product a practical solution for everyday use. Typically, manually raking a 2,500 sq. ft lot takes one person 2-3 hours to complete¹, much longer than the targeted time for the machine.

3. Leaf Disposal

a. Specification:

- i. The machine shall autonomously pick up leaves and dispose of them in its own storage unit.

b. Rationale:

- i. This eliminates the need for the user to repeatedly lift and carry weight, reducing the risk of neck, shoulder, and back injuries.

4. Battery Life

a. Specification:

- i. The machine's battery shall last 120 minutes and recharge in 50 minutes

b. Rationale:

- i. This ensures the battery lasts a 60-minute cleaning cycle, giving the machine extra buffer time in case it is a larger job or the conditions are unfavorable.²

5. Battery Replacement

a. Specification:

- i. The machine shall feature a swappable battery that can be replaced in under five minutes.

b. Rationale:

- i. Allows users to resume machine operation after a quick battery replacement. The swappable battery system keeps the machine working even if another battery from the last use is charging.

6. Storage

a. Specification:

- i. The machine shall have sufficient storage to accommodate at least 30 gallons of leaves with a weight capacity of 50 lbs.

¹ DeBruin, I. (2025, August 23). *How Long Does a Fall Yard Clean-Up Take?* - LambtonLawn LTD Blogs. LambtonLawn LTD Blogs. <https://lambtonlawn.com/blog/how-long-does-a-fall-yard-clean-up-take/>

- b. Rationale:
 - i. This storage capacity aligns closely with the capacity of a standard leaf bag, reducing the number of times the leaves need to be disposed of. This allows the machine to work more efficiently with minimum interruptions.
- 7. Compaction
 - a. Specification:
 - i. The machine shall compact collected leaves by a ratio of 1.5:1 in volume
 - b. Rationale:
 - i. This increases the machine's storage capacity, reducing the frequency with which leaves will have to be disposed of, and allows the machine to operate with minimal interruptions.
- 8. Battery Per Runtime
 - a. Specification:
 - i. The machine shall operate over a distance of 7,500 square feet in one charge, as according to conducted research, that is the average lawn size for suburban homes in the U.S.³
 - b. Rationale:
 - i. Improves reliability and eliminates mid-cycle cleaning, charging, or battery replacement.
- 9. Target Cost
 - a. Specification:
 - i. The machine shall not exceed the sale cost of \$550
 - b. Rationale:
 - i. Places the product competitively among existing products like leaf blowers and mowers, making it a financially viable option for homeowners. According to a survey, consumers are willing to spend an average of \$25 per cleaning. Assuming three cleanings per season and a machine life of eight seasons, the estimated value of the product is \$600, and \$550 is well below that.

³ Wasson, S. (2022, August 23). *The Average Yard Size by State and City*. Today's Homeowner. <https://todayshomeowner.com/lawn-garden/guides/average-yard-size/>

If there was an alternative investment for removing up leaves, how much would you pay per cleaning?

125 responses

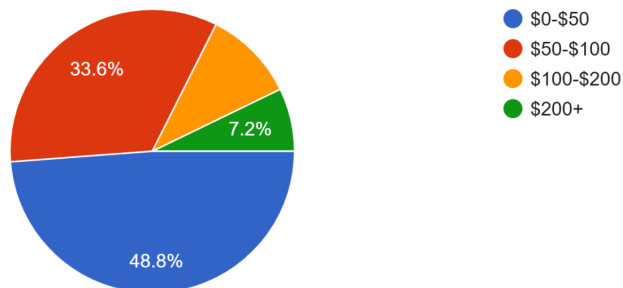


Figure C.1: Survey data for question “If there was an alternative investment for removing leaves, how much would you pay per cleaning”

Figure C.1 shows that approximately half of the surveyors would spend \$0- \$50 per leaf cleaning on a solution, and the rest would be willing to pay more.

Approximately how many times per season do you remove leaves from your yard in the fall?

127 responses

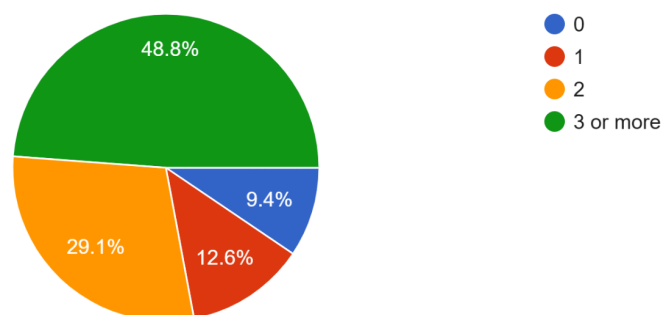


Figure C.2: Survey data for question “Approximately how many times per season do you remove leaves from your yard”

Figure C.2 shows that approximately half of the surveyors have to remove leaves from their yards 3 or more times.

10. Noise Level

a. Specification:

- i. The machine shall not exceed 60 dB during operation, measured at a distance of 5 feet.

b. Rationale:

- i. This ensures the machine is suitable for residential neighborhoods, reducing the disturbance neighbors or users may experience during operation.

11. Durability

a. Specification:

- i. The machine shall be operational and efficient for at least eight fall seasons.

b. Rationale:

- i. This ensures long-term value for users, making the product economically sensible for them. This lifespan is also on par with other lawncare devices, with most typically lasting 8-10 years⁴.

12. Ergonomics

a. Specification:

- i. Any human interference shall minimize awkward lifting positions and not exceed 20 pounds.

b. Rationale:

- i. This reduces the user's risk of injury, addressing the primary issue. This also makes it more convenient for the user to use the product.

13. Operating Environment

a. Specification:

- i. The machine shall operate effectively outdoors in temperatures ranging from 30 degrees Fahrenheit to 75 degrees Fahrenheit.

b. Rationale:

- i. This reflects typical fall weather conditions in the United States⁵, ensuring that the machine's performance during peak season is efficient and satisfactory.

14. Natural Conditions

a. Specification:

- i. The machine shall operate with resistance to conditions like dust, moisture, and uneven terrain

b. Rationale:

- i. This is necessary for outdoor operation, preventing wear and machine issues due to natural conditions. If not accounted for, the machine can face mechanical and electronic damage, leading to system crashes⁶.

15. Safety Precautions

a. Specification:

⁴Minnesota Equipment. (2023, July 12). *How Old Is Too Old for a Lawn Mower?* - Minnesota Equipment. Minnesota Equipment.

⁵ *Fall Temperature Averages for Each USA State - Current Results*. (n.d.). [www.currentresults.com](https://www.currentresults.com/Weather/US/average-state-temperatures-in-fall.php).
<https://www.currentresults.com/Weather/US/average-state-temperatures-in-fall.php>

⁶ Hulley, L. (2024). *The Impact of Dust in Electronics*. Bulgin.com.
<https://blog.bulgin.com/blog/the-impact-of-dust-in-electronics>

- i. The machine shall have an emergency stop in the event of malfunctions, debris clogging, or unexpected terrain.
- b. Rationale:
 - i. This provides immediate machine shutdown in dangerous situations, protecting users and their property from potential harm the machine can cause.

16. Maintenance

- a. Specification:
 - i. The machine's maintenance shall be limited to battery replacement and minimal inspection once per fall season.
- b. Rationale:
 - i. Limiting maintenance to only battery replacement and minor inspections reduces user responsibility and encourages continued use.

Conclusion

These design requirements make the solution unique and specify the product in a way that appeals to stakeholders. Focusing on an autonomous design will save time, minimize physical labor for homeowners, and improve efficiency for commercial lawn care services. These requirements ensure that the user has a reliable method of removing leaves during the fall for multiple seasons.